

# DISTINCT INTERSECTIONS IN PERMUTATION MATCHINGS

CLAUDE OPUS 5.5

ABSTRACT. For a permutation  $\pi$  of  $\{1, \dots, n\}$ , join  $(i, 0)$  to  $(\pi(i), 1)$  by a segment for each  $i$ , and let  $a(n)$  be the largest possible number of distinct points at which two or more of these segments meet. Trivially  $a(n) \leq \binom{n}{2}$ . We show that the defect  $D(n) = \binom{n}{2} - a(n)$  satisfies

$$\left(\frac{2}{3\sqrt{2}} - o(1)\right)n^{3/2} \leq D(n) \leq \left(\frac{\sqrt{3}}{2} + o(1)\right)n^{3/2}.$$

The lower bound comes from a one-dimensional variational problem for the number of points of the permutation graph on each slope-one diagonal. The upper bound comes from a blow-up of a modular hyperbola whose centre lies off the diagonal. This placement reduces the collinearities away from the diagonals to  $O(n \log^2 n)$ ; when the centre lies on the diagonal, they are of order  $n^{3/2}$ . We prove an exact identity that splits the defect into three nonnegative parts. It shows that the constant  $2/(3\sqrt{2})$  is the truth if and only if some permutations are nearly 321-avoiding, have an asymptotically triangular diagonal profile, and have few collinearities off the diagonals. Finally, an exhaustive search gives  $a(20) = 164$ .

An interactive companion to this paper, with live versions of its figures, is at  
<https://claude.ai/artifact/UMUAQFtaJFsUGa5vMLZpZ5>.

## 1. INTRODUCTION

For a permutation  $\pi \in S_n$ , draw the  $n$  segments

$$(i, 0) \longrightarrow (\pi(i), 1), \quad 1 \leq i \leq n,$$

between two parallel lines. Let  $A(\pi)$  be the number of distinct points at which at least two of the segments meet, and let

$$a(n) = \max_{\pi \in S_n} A(\pi).$$

The problem was posed by Tracy Morgan on Mathematics Stack Exchange [2], where the values up to  $n = 16$  are listed. Two segments  $i < j$  meet exactly when they cross, that is, when  $\pi(i) > \pi(j)$ ; such a pair is an *inversion* of  $\pi$ . Hence  $a(n) \leq \binom{n}{2}$ . The difficulty is that three or more segments may pass through one point. A cyclic shift  $\pi(i) \equiv i + m \pmod{n}$  has exactly  $m(n - m)$  intersection points, all distinct, so

$$\left\lfloor \frac{n^2}{4} \right\rfloor \leq a(n) \leq \binom{n}{2}.$$

These bounds differ by a factor of two. We determine the true order of the *defect*

$$D(n) = \binom{n}{2} - a(n),$$

and bound its leading constant on both sides.

**Theorem 1.1.** *As  $n \rightarrow \infty$ ,*

$$\left(\frac{2}{3\sqrt{2}} - o(1)\right)n^{3/2} \leq D(n) \leq \left(\frac{\sqrt{3}}{2} + o(1)\right)n^{3/2}.$$

*More precisely,  $D(n) \geq \frac{2}{3\sqrt{2}}n^{3/2} - O(n)$  for every  $n$ . Here  $\frac{2}{3\sqrt{2}} = 0.4714\dots$  and  $\frac{\sqrt{3}}{2} = 0.8660\dots$*

In particular  $a(n) = \frac{n^2}{2} - \Theta(n^{3/2})$ . The upper bound comes from an algebraic construction built on a modular hyperbola, and the position of the hyperbola's centre is decisive. With the centre on the diagonal, the same kind of construction has defect about  $1.4n^{3/2}$ , for a reason explained in [Remark 4.4](#).

We also compute the exact values for  $17 \leq n \leq 20$ .

**Theorem 1.2.**  $a(17) = 117$ ,  $a(18) = 133$ ,  $a(19) = 148$  and  $a(20) = 164$ .

The proof is a computer search described in [Section 6](#), together with explicit witnesses that can be checked by hand or by a few lines of code.

**1.1. Three costs.** Our results rest on an exact decomposition of the defect, stated in [Section 2](#). Write  $\rho(i) = n + 1 - \pi(i)$  for the reflected permutation. By [Lemma 2.1](#) below, the intersection points of  $\pi$  correspond to the lines of positive slope through two or more points of the graph  $\{(i, \rho(i))\}$ . A pair of points of this graph fails to produce a line of its own for one of three reasons:

- (i) the pair is an inversion of  $\rho$ , so its line has negative slope;
- (ii) the pair lies on a *slope-one diagonal* together with other points;
- (iii) the pair lies on some *other* positive-slope line together with other points.

Small inversion counts force the graph to stay close to the main diagonal. On the other hand, crowding the points onto few diagonals makes cost (ii) large. Balancing these two costs is a one-dimensional variational problem, whose optimum is a triangular profile of diagonal occupancies and whose value is  $\frac{2}{3\sqrt{2}}n^{3/2}$ . This gives the lower bound. Cost (iii) can be made negligible by algebraic constructions, but those constructions do not achieve the other two costs optimally: their inversions behave like those of a random permutation, and their diagonal occupancies take small integer values instead of following a smooth profile. This accounts for the gap between the two constants in [Theorem 1.1](#).

The next theorem makes this precise. For  $\rho \in S_n$  write  $d_i = \rho(i) - i$  for the *displacement* of  $i$ , let  $k_s = \#\{i : d_i = s\}$  be the number of points of the graph on the diagonal  $y = x + s$ , call  $k = (k_s)_{s \in \mathbb{Z}}$  the *profile* of  $\rho$ , and put

$$K_s = \left( \sqrt{n/2} - \frac{|s|}{2} \right)_+, \quad \Phi(k) = \frac{1}{2} \sum_s |s| k_s + \frac{1}{2} \sum_s k_s^2.$$

**Theorem 1.3.** *There are nonnegative quantities  $G(\rho)$ ,  $E_{\neq 1}(\rho)$ ,  $R(\rho)$  and a number  $\Phi_n^* = \frac{2}{3\sqrt{2}}n^{3/2} + O(n)$  such that, if  $\pi$  and  $\rho$  are related by  $\rho(i) = n + 1 - \pi(i)$ ,*

$$\binom{n}{2} - A(\pi) = G(\rho) + E_{\neq 1}(\rho) + (\Phi(k) - \Phi_n^*) + \Phi_n^* - R(\rho).$$

Here:

- $G(\rho) = \text{inv}(\rho) - \frac{1}{2} \sum_i |d_i|$ , where  $\text{inv}(\rho)$  is the number of inversions of  $\rho$ . It vanishes exactly when  $\rho$  avoids the pattern 321, that is, when there are no  $i < j < k$  with  $\rho(i) > \rho(j) > \rho(k)$ .
- $E_{\neq 1}(\rho) = \sum_{\ell} \left( \binom{t_{\ell}}{2} - 1 \right)$ , summed over the lines  $\ell$  of positive slope other than 1 that contain  $t_{\ell} \geq 2$  points of the graph of  $\rho$ .
- $\Phi(k) - \Phi_n^* \geq \frac{1}{2} \sum_s (k_s - K_s)^2$ .
- $0 \leq R(\rho) \leq \frac{3}{2}n$ .

**Corollary 1.4.**  $D(n) = \frac{2}{3\sqrt{2}}n^{3/2} + o(n^{3/2})$  if and only if there are permutations  $\rho_n \in S_n$  with

$$G(\rho_n), \quad E_{\neq 1}(\rho_n), \quad \Phi(k^{(n)}) - \Phi_n^*$$

all of order  $o(n^{3/2})$ , where  $k^{(n)}$  is the profile of  $\rho_n$ .

We conjecture that the lower bound is the truth.

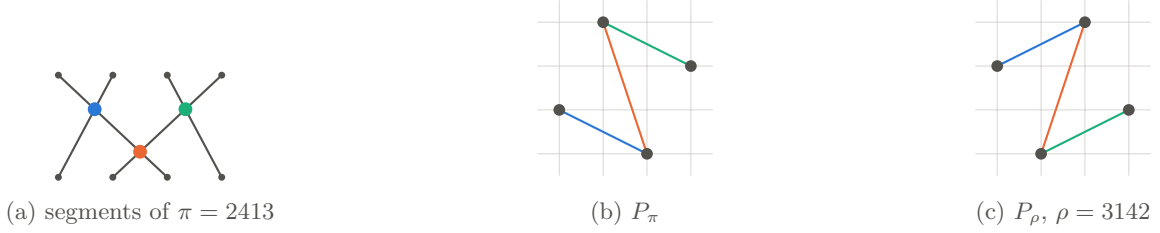


FIGURE 1. [Lemma 2.1](#) for  $\pi = 2413$ . Each intersection point in (a) corresponds to the line of the same colour in (b), which has negative slope and passes through the graph points of the segments meeting there. After the reflection  $\rho(i) = n + 1 - \pi(i)$ , these become the positive-slope lines in (c).

**Conjecture 1.5.**  $D(n) = \frac{2}{3\sqrt{2}}n^{3/2} + o(n^{3/2})$ .

[Section 5](#) discusses the evidence. Exact values for  $n \leq 20$  lie within one of a finite version of the lower bound ([Table 1](#)). At the same time, every construction we know pays for at least one of the three costs, and simulated annealing for  $n$  up to 800 does not come close to the conjectured constant. We do not know whether the conjecture is true.

**Organization.** [Section 2](#) translates the problem into secant lines and proves the decomposition. [Section 3](#) proves the lower bound and [Theorem 1.3](#). [Section 4](#) gives the construction behind the upper bound. [Section 5](#) discusses the sharp constant, and [Section 6](#) describes the computation of  $a(20)$ .

## 2. SECANTS AND THE DEFECT IDENTITY

For  $\pi \in S_n$  let  $P_\pi = \{(i, \pi(i)) : 1 \leq i \leq n\} \subset \mathbb{Z}^2$  be its graph. It is convenient to keep two planes apart: the *drawing plane*, in which the segments live between the lines  $y = 0$  and  $y = 1$ , and the *graph plane*, with coordinates  $(X, Y)$ , which contains  $P_\pi$ .

**Lemma 2.1.** *The distinct intersection points of the segments of  $\pi$  are in bijection with the lines of negative slope in the graph plane that contain at least two points of  $P_\pi$ . The segments through an intersection point are those indexed by the points of  $P_\pi$  on the corresponding line.*

*Proof.* At height  $t \in (0, 1)$  the segment of index  $i$  has horizontal coordinate  $(1 - t)i + t\pi(i)$ . Hence the segments  $i$  and  $j$  pass through the drawing point  $(x, t)$  exactly when  $(i, \pi(i))$  and  $(j, \pi(j))$  both lie on the graph-plane line

$$\ell_{x,t} : (1 - t)X + tY = x.$$

The map  $(x, t) \mapsto \ell_{x,t}$  is a bijection from  $\mathbb{R} \times (0, 1)$  to the lines of negative slope: the line  $bX + aY = c$  with  $a, b > 0$  corresponds to  $(x, t) = (c/(a + b), a/(a + b))$ . Under this bijection the segments through  $(x, t)$  correspond to the points of  $P_\pi$  on  $\ell_{x,t}$ .  $\square$

It is convenient to reflect the graph vertically. Put

$$\rho(i) = n + 1 - \pi(i).$$

Negative-slope lines through  $P_\pi$  become positive-slope lines through  $P_\rho$  ([Fig. 1](#)), so  $A(\pi)$  is the number of positive-slope lines containing at least two points of  $P_\rho$ . We write  $A(\rho)$  for this number when convenient.

A pair  $i < j$  lies on a positive-slope line exactly when  $\rho(i) < \rho(j)$ . The other pairs are the inversions of  $\rho$ , and they produce no intersection point. If a positive-slope line  $\ell$  contains  $t_\ell \geq 2$  points of  $P_\rho$ , its  $\binom{t_\ell}{2}$  pairs produce one line. Counting pairs therefore gives the *defect identity*

$$(2.1) \quad \binom{n}{2} - A(\rho) = \text{inv}(\rho) + \sum_{\ell} \left( \binom{t_\ell}{2} - 1 \right),$$

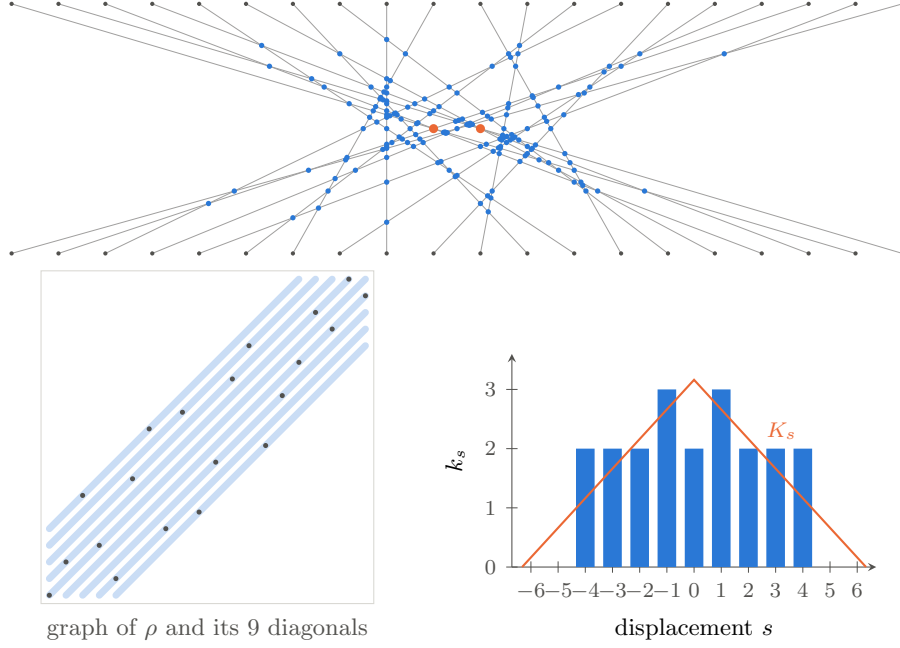


FIGURE 2. An optimal permutation for  $n = 20$ , found in Section 6:  $\pi = (20, 18, 14, 17, 19, 13, 10, 16, 9, 15, 12, 7, 5, 11, 8, 6, 3, 4, 1, 2)$ . Top: its segments and their  $164 = \binom{20}{2} - 26$  distinct intersection points. The two orange points are each crossed by three segments. Bottom left: the graph of  $\rho = 21 - \pi$ , whose points occupy the nine diagonals  $y = x + s$ ,  $|s| \leq 4$ . Bottom right: the occupancies  $k_s$  (bars) against the triangular profile  $K_s = (\sqrt{10} - |s|/2)_+$  that is optimal for large  $n$ .

where the sum is over positive-slope lines with  $t_\ell \geq 2$  and  $\text{inv}(\rho) = \#\{i < j : \rho(i) > \rho(j)\}$ .

**Diagonals.** Write  $d_i = \rho(i) - i$  for the *displacement* of  $i$  and  $k_s = \#\{i : d_i = s\}$  for the number of points on the *diagonal*  $y = x + s$ . We split the sum in (2.1) into the part  $E_1(\rho)$  coming from slope-one lines and the part  $E_{\neq 1}(\rho)$  coming from all other positive-slope lines. With

$$h(k) = \begin{cases} 0, & k \leq 1, \\ \binom{k}{2} - 1, & k \geq 2, \end{cases}$$

we have  $E_1(\rho) = \sum_s h(k_s)$ , and the defect identity reads

$$(2.2) \quad \binom{n}{2} - A(\rho) = \text{inv}(\rho) + \sum_s h(k_s) + E_{\neq 1}(\rho).$$

A shear makes the diagonals horizontal. The map  $(x, y) \mapsto (x, y - x)$  sends  $P_\rho$  to the set  $\{(i, d_i)\}$ , which lies in the horizontal strip  $|y| \leq \max |d_i|$ . It sends slope-one lines to horizontal lines and preserves collinearity. We use these *sheared coordinates* in Section 4. Figure 2 shows an optimal permutation for  $n = 20$ , both as segments and as a graph with its diagonals.

### 3. THE LOWER BOUND

The first two costs in (2.2) are both controlled by the displacements. Large displacements force inversions, and small displacements crowd many points onto few diagonals.

The quantity  $\sum_i |\rho(i) - i|$  is known as *Spearman's footrule*.

**Lemma 3.1** (Diaconis–Graham [1]). *For every  $\rho \in S_n$ ,*

$$\text{inv}(\rho) \geq \frac{1}{2} \sum_{i=1}^n |\rho(i) - i| = \frac{1}{2} \sum_s |s|k_s.$$

*Equality holds if and only if  $\rho$  avoids the pattern 321.*

*Proof.* Sort  $\rho$  to the identity by adjacent transpositions of inverted neighbours. This takes exactly  $\text{inv}(\rho)$  steps, each changing  $\sum_i |\rho(i) - i|$  by at most 2. The equality case is due to Petersen and Tenner [3]. They define the depth of a permutation, show that it equals  $\frac{1}{2} \sum_i |\rho(i) - i|$ , and prove that it equals the number of inversions exactly for 321-avoiding permutations.  $\square$

We call  $G(\rho) = \text{inv}(\rho) - \frac{1}{2} \sum_i |d_i| \geq 0$  the *footrule gap*. The diagonal term is handled by completing a square.

**Lemma 3.2.** *Let  $\lambda > 0$ , and let  $(k_s)_{s \in \mathbb{Z}}$  be nonnegative reals with  $\sum_s k_s = n$ . Put  $K_s = (\lambda - |s|/2)_+$ . Then*

$$\Phi(k) = \frac{1}{2} \sum_s |s|k_s + \frac{1}{2} \sum_s k_s^2 \geq \lambda n - \frac{1}{2} \sum_s K_s^2 + \frac{1}{2} \sum_s (k_s - K_s)^2.$$

*Proof.* Since  $\sum_s k_s = n$ , we have  $\Phi(k) = \lambda n + \sum_s (\frac{1}{2}k_s^2 - a_s k_s)$  with  $a_s = \lambda - |s|/2$ . If  $a_s > 0$  then  $\frac{1}{2}k_s^2 - a_s k_s = \frac{1}{2}(k_s - K_s)^2 - \frac{1}{2}K_s^2$ . If  $a_s \leq 0$  then  $K_s = 0$  and  $\frac{1}{2}k_s^2 - a_s k_s \geq \frac{1}{2}k_s^2 = \frac{1}{2}(k_s - K_s)^2 - \frac{1}{2}K_s^2$ . Summing over  $s$  gives the claim.  $\square$

Take  $\lambda = \sqrt{n/2}$  and define  $\Phi_n^* = \lambda n - \frac{1}{2} \sum_s K_s^2$ . Comparing the sum with an integral gives  $\sum_s K_s^2 = \frac{4}{3}\lambda^3 + O(\lambda^2)$ , and hence

$$(3.1) \quad \Phi_n^* = \lambda n - \frac{2}{3}\lambda^3 + O(n) = \frac{4}{3}\left(\frac{n}{2}\right)^{3/2} + O(n) = \frac{2}{3\sqrt{2}}n^{3/2} + O(n).$$

*Proof of Theorem 1.3.* For  $k \geq 2$  we have  $h(k) = \frac{1}{2}k^2 - \frac{1}{2}k - 1$ , while  $h(0) = h(1) = 0$ . Hence  $\sum_s h(k_s) = \frac{1}{2} \sum_s k_s^2 - R(\rho)$  with  $0 \leq R(\rho) \leq \sum_{s:k_s \geq 1} (\frac{1}{2}k_s + 1) \leq \frac{3}{2}n$ . Substituting this and  $\text{inv}(\rho) = G(\rho) + \frac{1}{2} \sum_s |s|k_s$  into (2.2) gives

$$\binom{n}{2} - A(\rho) = G(\rho) + E_{\neq 1}(\rho) + \Phi(k) - R(\rho),$$

which is the stated identity. The properties of  $G$  come from Lemma 3.1, the bound on  $\Phi(k) - \Phi_n^*$  comes from Lemma 3.2, and  $\Phi_n^*$  was computed in (3.1).  $\square$

*Proof of the lower bound in Theorem 1.1.* In the identity of Theorem 1.3, the first three terms are nonnegative and  $R \leq \frac{3}{2}n$ . So every  $\pi \in S_n$  has defect at least  $\Phi_n^* - \frac{3}{2}n = \frac{2}{3\sqrt{2}}n^{3/2} - O(n)$ .  $\square$

*Proof of Corollary 1.4.* If  $G$ ,  $E_{\neq 1}$  and  $\Phi - \Phi_n^*$  are all  $o(n^{3/2})$  for some  $\rho_n$ , the identity gives  $D(n) \leq \Phi_n^* + o(n^{3/2})$ . Conversely, suppose  $D(n) = \Phi_n^* + o(n^{3/2})$  and let  $\rho_n$  be optimal. The identity gives

$$G(\rho_n) + E_{\neq 1}(\rho_n) + (\Phi(k^{(n)}) - \Phi_n^*) = D(n) - \Phi_n^* + R(\rho_n) = o(n^{3/2}),$$

and all three terms on the left are nonnegative.  $\square$

| $n$    | 1 | 2 | 3 | 4 | 5 | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20         |
|--------|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|------------|
| $a(n)$ | 0 | 1 | 2 | 5 | 8 | 13 | 17 | 23 | 30 | 39 | 47 | 57 | 67 | 79 | 90 | 103 | 117 | 133 | 148 | <b>164</b> |
| $D(n)$ | 0 | 0 | 1 | 1 | 2 | 2  | 4  | 5  | 6  | 6  | 8  | 9  | 11 | 12 | 15 | 17  | 19  | 20  | 23  | 26         |
| $L(n)$ | 0 | 0 | 1 | 1 | 2 | 2  | 3  | 4  | 5  | 6  | 8  | 9  | 11 | 12 | 14 | 16  | 18  | 20  | 22  | 25         |

TABLE 1. Exact values of  $a(n)$  and  $D(n) = \binom{n}{2} - a(n)$ , and the lower bound  $L(n)$ . Values for  $n \leq 16$  are from [2]; values for  $17 \leq n \leq 20$  are computed in Section 6.

**The finite form of the bound.** The same argument gives a lower bound that can be evaluated exactly for each  $n$ . Discarding  $E_{\neq 1}$  in (2.2) and applying Lemma 3.1 shows

$$D(n) \geq L(n) := \min_k \left[ \frac{1}{2} \sum_s |s| k_s + \sum_s h(k_s) \right],$$

the minimum being over nonnegative integer vectors  $(k_s)$  with  $\sum_s k_s = n$ . The function  $k \mapsto \frac{1}{2}|s|k + h(k)$  has nondecreasing increments  $\frac{1}{2}|s|, \frac{1}{2}|s|, \frac{1}{2}|s|+2, \frac{1}{2}|s|+3, \dots$ , so the minimum is computed greedily by taking the  $n$  smallest increments over all  $s$ . Table 1 compares  $L(n)$  with the exact values. The bound is attained for  $n \leq 6$ , for  $10 \leq n \leq 14$  and for  $n = 18$ , and is off by one otherwise.

#### 4. THE CONSTRUCTION

We build permutations by repeating one permutation  $g$  of  $\mathbb{F}_p$  along the diagonal and appending a short identity tail. Throughout this section we use the sheared coordinates of Section 2, in which the graph point  $(x, \rho(x))$  becomes  $(x, \rho(x) - x)$ . We call the second coordinate the *height* of the point, so slope-one lines become lines of constant height. Three things must be controlled: the inversions of  $g$ , how many of its points share each displacement, and the lines of  $\mathbb{F}_p^2$  that contain three points of its graph.

**4.1. The block.** Let  $p \geq 5$  be prime and  $c = \frac{p-1}{2}$ . Define  $g : \mathbb{F}_p \rightarrow \mathbb{F}_p$  by

$$g(0) = c, \quad g(x) = c + x^{-1} \quad (x \neq 0).$$

Since  $x \mapsto c + x^{-1}$  is a bijection from  $\mathbb{F}_p^\times$  to  $\mathbb{F}_p \setminus \{c\}$ , the map  $g$  is a permutation. We identify  $\mathbb{F}_p$  with  $\{0, 1, \dots, p-1\}$  whenever we need integers. The graph of  $g$  in  $\mathbb{F}_p^2$  consists of the modular hyperbola  $(y - c)x = 1$  together with its centre  $(0, c)$ . Figure 3 compares it with the block  $x \mapsto x/(x-1)$ , whose centre lies on the diagonal.

**Lemma 4.1.** *Every line of  $\mathbb{F}_p^2$  meets the graph  $\Gamma$  of  $g$  in at most three points. The lines that meet it in three points are exactly the lines through the centre  $(0, c)$  and the two points with first coordinates  $t$  and  $-t$ , for  $t \in \mathbb{F}_p^\times$ .*

*Proof.* A line meets the nondegenerate conic  $(y - c)x = 1$  in at most two points, so it meets  $\Gamma$  in at most three, and in three only if it passes through the centre. The line through the centre and  $(t, c + t^{-1})$  is  $y - c = t^{-2}x$ . It meets the conic where  $t^{-2}x^2 = 1$ , that is, at  $x = \pm t$ .  $\square$

For  $0 \leq r < p$  let  $d_r = g(r) - r \in (-p, p)$  be the integer displacement, and let  $m_e = \#\{r : d_r = e\}$  be the number of block points on the diagonal  $y = x + e$ .

**Lemma 4.2.** *For  $1 \leq t \leq p-1$  we have  $g(t) + g(p-t) = p-1$  and  $d_t + d_{p-t} = -1$ . Moreover  $d_0 = c$  and  $m_e \leq 3$  for every  $e$ .*

*Proof.* We have  $g(t) \equiv c + t^{-1}$  and  $g(p-t) \equiv c - t^{-1}$ , so  $g(t) + g(p-t) \equiv 2c = p-1 \pmod{p}$ . Both values lie in  $[0, p-1]$ , so the sum is  $p-1$  exactly. Subtracting  $t + (p-t) = p$  gives the second identity. Finally  $d_r = e$  forces  $g(r) - r \equiv e$ ; for  $r \neq 0$  this is the quadratic  $r^2 + (e - c)r - 1 \equiv 0$ , which has at most two roots, and  $r = 0$  adds at most one more.  $\square$

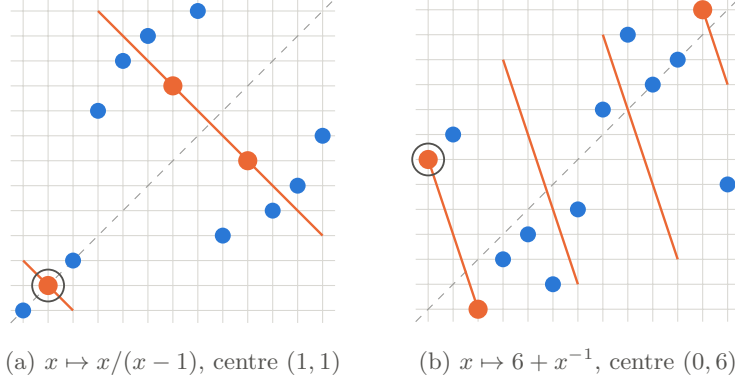


FIGURE 3. Two blocks for  $p = 13$ . Each graph is a modular hyperbola plus its centre (ringed). Only lines through the centre contain three points; one such line is drawn, wrapping around the torus. In (a) the centre lies on the diagonal. The hyperbola is symmetric about its centre, so the highlighted triple has displacements  $0, 3, -3$ , and its copies in the blow-up of [Section 4.2](#) line up often. In (b) the displacements of the triple are  $c, e, -1 - e$  with  $c = 6$  and  $e = -2$  ([Lemma 4.2](#)), and its copies rarely line up.

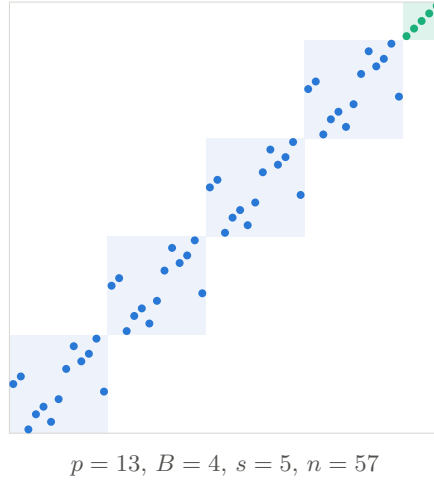


FIGURE 4. The permutation of [Section 4.2](#):  $B$  copies of the block of [Fig. 3\(b\)](#) along the diagonal, followed by an identity tail of length  $s < p$ .

**4.2. The blow-up.** Let  $B \geq 1$  and  $0 \leq s < p$ , and put  $n = Bp + s$ . Define  $\rho \in S_n$ , on  $\{0, \dots, n-1\}$ , by

$$\rho(bp + r) = bp + g(r) \quad (0 \leq b < B, 0 \leq r < p), \quad \rho(x) = x \quad (Bp \leq x < n).$$

The graph consists of  $B$  diagonal copies of the block and an identity tail of length  $s$  ([Fig. 4](#)). In sheared coordinates the *block points* are  $(bp + r, d_r)$ , and we call  $r$  the *residue* of such a point. The *tail points* are  $(x, 0)$  with  $Bp \leq x < n$ . A line in sheared coordinates is *horizontal* if it corresponds to a slope-one line.

We bound the three costs in [\(2.2\)](#) in turn.

**Lemma 4.3** (Collinearities among block points). *A non-horizontal line contains at most three block points. The number of lines containing three block points is at most  $4B^2 + pB$ . Hence the block points contribute at most  $8B^2 + 2pB$  to  $E_{\neq 1}(\rho)$ .*

*Proof.* Two block points with the same residue  $r$  have the same height  $d_r$ , so a non-horizontal line contains at most one block point from each residue class. If block points with distinct

residues  $r_1, r_2, r_3$  lie on one line of  $\mathbb{Z}^2$ , then reducing the collinearity determinant modulo  $p$  shows that the points  $(r_i, d_{r_i} \bmod p)$  are collinear in  $\mathbb{F}_p^2$ . These points lie on the image of  $\Gamma$  under the shear  $(x, y) \mapsto (x, y - x)$ , which preserves lines. By [Lemma 4.1](#) there are at most three of them, and three only if the residues are  $0, t, p - t$  for some  $t$ .

Fix  $t$  and let  $E_1 = d_t - c$  and  $E_2 = d_{p-t} - c$ . By [Lemma 4.2](#),  $E_2 = -(E_1 + p)$ . If  $E_1 = 0$  or  $E_1 = -p$ , the centre is at the same height as one of the other two points, and a non-horizontal line cannot contain all three. Otherwise  $\gcd(E_1, E_1 + p) = \gcd(E_1, p) = 1$  and  $\max(|E_1|, |E_1 + p|) \geq p/2$ . Three block points  $(x_0, c), (x_1, d_t), (x_2, d_{p-t})$  with  $x_0 \equiv 0, x_1 \equiv t, x_2 \equiv -t \pmod{p}$  are collinear if and only if  $(x_1 - x_0)E_2 = (x_2 - x_0)E_1$ . By coprimality this means

$$x_1 - x_0 = E_1\mu, \quad x_2 - x_0 = -(E_1 + p)\mu$$

for some integer  $\mu$ . The congruence  $x_1 \equiv t$  fixes  $\mu$  modulo  $p$ , and  $|x_1 - x_0|, |x_2 - x_0| < Bp$  gives  $|\mu| < 2B$ . So for each of the  $B$  choices of  $x_0$  there are at most  $4B/p + 1$  choices of  $\mu$ . Summing over  $t$  shows that there are at most  $(p - 1)B(4B/p + 1) \leq 4B^2 + pB$  lines. Each line with three points contributes  $\binom{3}{2} - 1 = 2$  to  $E_{\neq 1}$ .  $\square$

*Remark 4.4.* The coprimality of  $E_1$  and  $E_2$  is the whole point of placing the centre at height  $c = \frac{p-1}{2}$ . For comparison, the block  $x \mapsto x/(x - 1)$  has its centre  $(1, 1)$  on the diagonal ([Fig. 3\(a\)](#)). A line through the centre of a hyperbola meets it in two points that are symmetric about the centre. For most such lines the integer representatives respect this symmetry, so the outer points have displacements  $\delta$  and  $-\delta$  exactly, while the centre has displacement 0. Then  $E_2 = -E_1$ , the condition above becomes  $x_1 + x_2 = 2x_0$ , and each such triple lifts to about  $B^2$  lines. The collinear excess off the diagonals is then of order  $pB^2$ , comparable to  $n^{3/2}$ . Numerically it is about  $\frac{1}{2}nB$ . With the best block count that construction has defect about  $1.37 n^{3/2}$  ([Table 2](#)).

**Lemma 4.5** (Collinearities with the tail). *The lines containing a tail point contribute  $O(sp + Bp \log^2 p)$  to  $E_{\neq 1}(\rho)$ .*

*Proof.* Tail points all have height 0, so a non-horizontal line contains at most one tail point. By the argument of [Lemma 4.3](#) it contains at most three block points, so it contributes at most  $\binom{4}{2} - 1 = 5$  to  $E_{\neq 1}$ . It is therefore enough to bound the number  $N$  of triples  $(w, y, z)$  where  $w$  is a tail point and  $y, z$  are block points on a common non-horizontal line through  $w$ .

Fix a tail point  $w = (x_w, 0)$  and a block point  $y = (x_y, \alpha)$  of residue  $r_y$ , with  $\alpha = d_{r_y} \neq 0$  (if  $\alpha = 0$  the line  $wy$  is horizontal). As in [Lemma 4.3](#), a block point  $z$  on the line  $wy$  has a residue  $r_z$  such that  $(x_w, 0), (r_y, \alpha)$  and  $(r_z, d_{r_z})$  are collinear modulo  $p$ . The line through the first two points meets the sheared graph of  $g$  in at most three points, one of which is  $(r_y, \alpha)$ , so there are at most two such  $r_z$ . For a given  $r_z$ , write  $\beta = d_{r_z}$ . The line  $wy$  passes through a lattice point at height  $\beta$  exactly when  $\alpha$  divides  $(x_y - x_w)\beta$ , that is, when  $\alpha' = |\alpha|/\gcd(\alpha, \beta)$  divides  $x_y - x_w$ . As the block containing  $y$  varies,  $x_y - x_w$  runs through an arithmetic progression of difference  $p$ , and  $\alpha'$  is coprime to  $p$  because  $0 < \alpha' < p$ . So at most  $B/\alpha' + 1$  blocks give a point  $z$ . Hence

$$N \leq \sum_w \sum_{r_y} \sum_{r_z} \left( \frac{B \gcd(\alpha, \beta)}{|\alpha|} + 1 \right).$$

Now fix  $r_y$  and let  $w$  vary over the  $s < p$  tail points, whose first coordinates are distinct modulo  $p$ . The lines through  $(r_y, \alpha)$  and the points  $(x_w, 0)$  are distinct lines of  $\mathbb{F}_p^2$ , because  $\alpha \not\equiv 0$ . Each point of the sheared graph lies on only one line through  $(r_y, \alpha)$ . So the residues

$r_z$  arising for different  $w$  are distinct, and

$$\begin{aligned} \sum_w \sum_{r_z} \gcd(\alpha, \beta) &\leq \sum_{r=0}^{p-1} \gcd(\alpha, d_r) \leq 3 \sum_{|e| < p} \gcd(\alpha, e) \\ &\leq 3 \sum_{u|\alpha} \varphi(u) \left( \frac{2p}{u} + 1 \right) \leq 6p \tau(|\alpha|) + 3|\alpha|, \end{aligned}$$

using  $m_e \leq 3$  and  $\gcd(\alpha, e) = \sum_{u|\gcd(\alpha, e)} \varphi(u)$ ; here  $\varphi$  is Euler's function and  $\tau$  counts divisors. The nonzero values  $\alpha = d_{r_y}$  each occur at most three times, so

$$N \leq 2sp + \sum_{r_y} \left( \frac{6Bp \tau(|\alpha|)}{|\alpha|} + 3B \right) \leq 2sp + 3Bp + 36Bp \sum_{k < p} \frac{\tau(k)}{k} = O(sp + Bp \log^2 p). \quad \square$$

The inversions and the diagonal occupancies of the block depend on how the points  $(x, x^{-1})$  are distributed in  $[0, p]^2$ . We use the classical consequence of Weil's bound for Kloosterman sums [5]; see Shparlinski's survey [4] for this and many refinements.

**Lemma 4.6.** *For integer intervals  $I, J \subseteq [0, p)$ ,*

$$\#\{x \in I \setminus \{0\} : x^{-1} \bmod p \in J\} = \frac{|I||J|}{p} + O(\sqrt{p} \log^2 p).$$

**Lemma 4.7.**  *$\text{inv}(g) = \frac{p^2}{4} + O(p^{3/2} \log^2 p)$  and  $\sum_e m_e^2 = \frac{3p}{2} + O(\sqrt{p} \log^2 p)$ .*

*Proof.* Since  $g(x) \equiv c + x^{-1}$ , the condition  $g(x) \in J$  for an interval  $J$  is a condition  $x^{-1} \in J - c$ , where  $J - c$  is a union of at most two intervals modulo  $p$ . So Lemma 4.6 applies to all counts below, and  $x = 0$  changes each count by at most one.

For the inversions, write  $\text{inv}(g) = \sum_x \#\{y > x : g(y) < g(x)\} = \sum_x ((p-1-x)g(x)/p + O(\sqrt{p} \log^2 p))$ . Next,  $\sum_x g(x) = \binom{p}{2}$ , and

$$\begin{aligned} \sum_x xg(x) &= \sum_{0 \leq a, b < p} \#\{x > a : g(x) > b\} = \sum_{a, b} \left( \frac{(p-1-a)(p-1-b)}{p} + O(\sqrt{p} \log^2 p) \right) \\ &= \frac{p(p-1)^2}{4} + O(p^{5/2} \log^2 p). \end{aligned}$$

Hence  $\sum_x (p-1-x)g(x)/p = \frac{(p-1)^2}{4} + O(p^{3/2} \log^2 p)$ , which gives the first claim.

For the second claim,  $\sum_e m_e^2$  counts ordered pairs  $(r, r')$  with  $d_r = d_{r'}$ . The diagonal pairs  $r = r'$  contribute  $p$ . Consider  $r \neq r'$ . For nonzero  $r, r'$ , the congruence  $d_r \equiv d_{r'}$  reads  $r^{-1} - r \equiv r'^{-1} - r'$ , which is equivalent to  $rr' \equiv -1$ . For  $r = 0$  it reads  $r' = \pm 1$ . Among the pairs involving 0, only  $\{0, 1\}$  has equal integer displacements, since  $d_0 = d_1 = c$  and  $d_{p-1} = c - p$ .

Now let  $r \in [1, p-1]$  with  $r^2 \not\equiv -1$ , and put  $u = r$ ,  $v = r^{-1} \bmod p$  and  $r' = p - v$ . Then  $g(r) = c + v - p[v > c]$  and  $g(r') = c - u + p[u > c]$ , so

$$d_r = c + v - u - p[v > c], \quad d_{r'} = c + v - u - p + p[u > c].$$

These are equal exactly when exactly one of  $u > c$  and  $v > c$  holds. By Lemma 4.6 this happens for  $\frac{p-1}{2} + O(\sqrt{p} \log^2 p)$  values of  $r$ . Altogether  $\sum_e m_e^2 = p + 2 + \frac{p}{2} + O(\sqrt{p} \log^2 p)$ .  $\square$

### 4.3. Proof of the upper bound.

**Theorem 4.8.** *With  $\rho$  as in Section 4.2,*

$$\binom{n}{2} - A(\rho) \leq \frac{Bp^2}{4} + \frac{3B^2p}{4} + O(Bp^{3/2} \log^2 p + B^2 \sqrt{p} \log^2 p + sp + Bp \log^2 p).$$

Consequently  $D(n) \leq (\frac{\sqrt{3}}{2} + o(1))n^{3/2}$ .

*Proof.* We bound the three terms of (2.2).

*Inversions.* The blocks and the tail form a direct sum, so  $\text{inv}(\rho) = B \text{inv}(g) = \frac{Bp^2}{4} + O(Bp^{3/2} \log^2 p)$  by Lemma 4.7.

*Diagonals.* The diagonal  $y = x + e$  contains  $k_e = Bm_e + s[e = 0]$  points. With  $h(k) \leq k^2/2$ ,  $m_0 \leq 3$  and Lemma 4.7,

$$\sum_e h(k_e) \leq \frac{1}{2} \sum_e k_e^2 \leq \frac{B^2}{2} \sum_e m_e^2 + 3Bs + \frac{s^2}{2} = \frac{3B^2p}{4} + O(B^2\sqrt{p} \log^2 p + Bs + s^2).$$

*Other lines.* By Lemmas 4.3 and 4.5,  $E_{\neq 1}(\rho) = O(B^2 + pB + sp + Bp \log^2 p)$ .

For the asymptotic statement, fix  $\varepsilon > 0$ . For large  $n$  the prime number theorem provides a prime  $p$  with  $(1 - \varepsilon)\sqrt{3n} \leq p \leq \sqrt{3n}$ . Put  $B = \lfloor n/p \rfloor$  and  $s = n - Bp < p$ . Then  $Bp^2/4 \leq np/4 \leq \frac{\sqrt{3}}{4}n^{3/2}$  and

$$\frac{3B^2p}{4} \leq \frac{3n^2}{4p} \leq \frac{\sqrt{3}}{4}(1 - \varepsilon)^{-1}n^{3/2}.$$

Since  $B \asymp p \asymp \sqrt{n}$ , every error term is  $O(n^{5/4} \log^2 n)$ . Hence  $D(n) \leq \frac{\sqrt{3}}{2}(1 - \varepsilon)^{-1}n^{3/2} + O(n^{5/4} \log^2 n)$ , and  $\varepsilon$  was arbitrary.  $\square$

*Proof of Theorem 1.1.* The lower bound was proved in Section 3, and the upper bound is Theorem 4.8.  $\square$

The choice  $p \approx \sqrt{3n}$ , that is  $B \approx p/3$ , balances the inversion term  $\frac{1}{4}Bp^2$  against the diagonal term  $\frac{3}{4}B^2p$ . Table 2 lists the exact defects of the construction. The three terms behave as predicted:  $\text{inv}(g)/p^2$  ranges from 0.23 to 0.26,  $\sum_e m_e^2/p$  from 1.46 to 1.54, and  $E_{\neq 1}$  is at most  $2.4n$ . The ratio  $D/n^{3/2}$  decreases slowly towards  $\frac{\sqrt{3}}{2}$ .

## 5. THE SHARP CONSTANT

By Corollary 1.4, Conjecture 1.5 holds if and only if the three costs  $G$ ,  $E_{\neq 1}$  and  $\Phi - \Phi^*$  can be made  $o(n^{3/2})$  simultaneously. Each of them can be made small on its own. The difficulty is to make them small together, and the known constructions illustrate this.

**What the block construction pays.** A heuristic comparison explains both constants. Consider a blow-up of a block of size  $p$  with  $B$  copies, as in Section 4.2, and imagine an idealized block whose displacements are spread like the difference of two independent uniform variables on  $[0, p)$ . If its inversions equalled half its footprint, as they do for a 321-avoiding permutation, they would number about  $\frac{1}{6}p^2$ . If moreover the occupancies  $Bm_e$  followed the continuous profile  $B(1 - |e|/p)$ , the diagonal term would be  $\frac{1}{2}B^2 \cdot \frac{2}{3}p$ . The two costs  $\frac{1}{6}Bp^2 + \frac{1}{3}B^2p$  are minimized at  $B = p/2$ , where their sum is exactly  $\frac{2}{3\sqrt{2}}n^{3/2}$ . The modular hyperbola replaces  $\frac{1}{6}$  by  $\frac{1}{4}$ , since its inversions behave like those of a random permutation. It replaces  $\frac{1}{3}$  by  $\frac{3}{4}$ , since its occupancies take the integer values 0, 1, 2, 3. The resulting optimum is  $\frac{\sqrt{3}}{2}n^{3/2}$ . In exchange, the algebraic structure makes  $E_{\neq 1}$  negligible.

No single block can realize the idealized values. In a 321-avoiding permutation of  $\{0, \dots, p-1\}$ , the points above the diagonal form an increasing sequence. If one of them is  $(x_0, y_0)$ , every other one lies either to the lower left of it or to the upper right, so there are at most  $x_0 + (p-1-y_0) + 1 = p - (y_0 - x_0)$  of them. The idealized block would have about  $p/2$  points above the diagonal and displacements close to  $p$ . So a construction attaining Conjecture 1.5, if one exists, cannot be a blow-up of a single block.

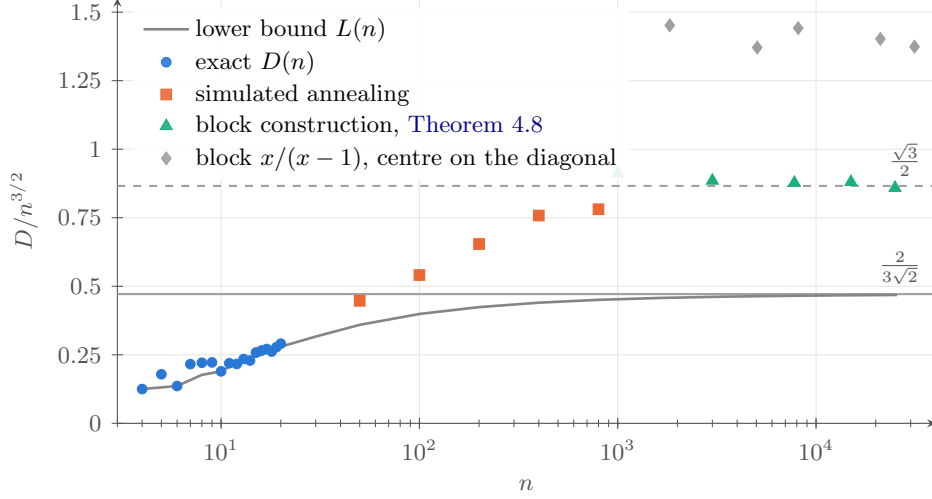


FIGURE 5. The normalized defect  $D/n^{3/2}$ . The exact values for  $n \leq 20$  (blue) follow the finite lower bound  $L(n)$  (grey curve), which tends to  $\frac{2}{3\sqrt{2}}$ . The block construction of Theorem 4.8 (aqua) tends to  $\frac{\sqrt{3}}{2}$ . Annealing (orange) beats both constructions for moderate  $n$ . The numbers are listed in Table 2.

**A periodic variant.** The block structure forces displacements in  $(-p, p)$  with a triangular distribution. A periodic construction removes this constraint. Take  $\phi(x) = x + F(x \bmod p)$ , where  $F(r)$  is the representative of  $g(r) - r$  in  $(-p/2, p/2)$ , and let  $\rho$  be the permutation of  $\{0, \dots, n-1\}$  with the same relative order as  $\phi$ . Heuristically this gives  $\frac{1}{6}p^2$  inversions per period and  $\sum_e m_e^2 \approx 2p$ , for a constant of  $\sqrt{2/3} \approx 0.816$  at  $p \approx \sqrt{6n}$ . The exact defects for  $n \leq 20,000$  lie between 0.82 and 0.86 times  $n^{3/2}$ , with  $E_{\neq 1}$  below  $5.1n$ . The obstacle to a proof is the  $O(p)$  points near the two ends of  $[0, n)$ , where  $\rho$  differs from  $\phi$ . The averaging argument of Lemma 4.5 needs those points to share a common height, and here they do not.

**What randomness pays.** At the other extreme, a uniformly random 321-avoiding permutation has  $G = 0$ . Its displacements are of order  $\sqrt{n}$ , so the diagonal term is of the right order. However, its collinear excess off the diagonals grows faster than  $n^{3/2}$ . In our computations  $E_{\neq 1}/n^{3/2}$  increased from 0.64 at  $n = 500$  to 1.08 at  $n = 16,000$ , consistent with growth like  $n^{3/2} \log n$ . The two increasing chains that make up such a permutation take small steps and therefore produce many near-diagonal collinearities.

**Local search.** Simulated annealing, with moves that swap two nearby values, finds permutations better than both constructions for moderate  $n$ :  $0.447 n^{3/2}$  at  $n = 50$  and  $0.541 n^{3/2}$  at  $n = 100$  (Fig. 5). The advantage shrinks as  $n$  grows. At  $n = 800$ , annealing started from the periodic construction improves it from 0.843 to 0.781. These values are upper bounds for  $D(n)$  and say nothing about the limit. They do show that none of the three costs is negligible in any permutation we have found.

#### Open problems.

- (1) Prove or disprove Conjecture 1.5. By Corollary 1.4, a proof requires permutations that are nearly 321-avoiding, have a nearly triangular profile, and have  $o(n^{3/2})$  collinear excess off the diagonals. A disproof requires showing that these three properties are incompatible.
- (2) Make the periodic variant rigorous, which would give  $D(n) \leq (\sqrt{2/3} + o(1))n^{3/2}$ .
- (3) Find any construction with an explicit constant below the ones above, for instance one whose inversions are not those of a random permutation.

| construction                       | $n$   | $p$ | $D/n^{3/2}$ | $\text{inv}/n^{3/2}$ | $E_1/n^{3/2}$ | $E_{\neq 1}/n$ |
|------------------------------------|-------|-----|-------------|----------------------|---------------|----------------|
| block, <a href="#">Theorem 4.8</a> | 1000  | 53  | 0.912       | 0.372                | 0.474         | 2.13           |
|                                    | 3001  | 89  | 0.885       | 0.380                | 0.463         | 2.34           |
|                                    | 7777  | 151 | 0.877       | 0.417                | 0.436         | 2.15           |
|                                    | 15000 | 211 | 0.880       | 0.455                | 0.420         | 0.68           |
|                                    | 25013 | 271 | 0.859       | 0.400                | 0.448         | 1.78           |
| periodic variant                   | 5000  | 173 | 0.824       | 0.380                | 0.391         | 3.69           |
|                                    | 20000 | 347 | 0.842       | 0.407                | 0.399         | 5.10           |
| block $x/(x-1)$                    | 5050  | 101 | 1.371       | 0.330                | 0.700         | 24.26          |
|                                    | 31375 | 251 | 1.374       | 0.318                | 0.707         | 61.75          |

TABLE 2. Exact defects of the constructions, split as in (2.2). The last column shows that  $E_{\neq 1}$  is  $O(n)$  for the block of [Theorem 4.8](#) and grows like  $nB$  for the block  $x/(x-1)$ , which is shown with  $B \approx p/2$ .

- (4) Improve the lower bound by showing that  $G + E_{\neq 1}$  must be of order  $n^{3/2}$  whenever the profile is nearly triangular.

## 6. THE EXACT VALUE OF $a(20)$

Write  $D(\rho) = \binom{n}{2} - A(\rho)$  for the defect of  $\rho \in S_n$ ; the search minimizes it using the decomposition (2.2). By [Lemma 3.1](#) we have  $D(\rho) \geq \mathcal{E}(k) = \frac{1}{2} \sum_s |s|k_s + \sum_s h(k_s)$ , which depends only on the displacement profile  $k$ . To decide whether some  $\rho$  has  $D(\rho) \leq D_{\max}$ , we proceed as follows.

- (1) Enumerate the profiles  $k$  with  $\sum_s k_s = n$ ,  $\sum_s s k_s = 0$  and  $\mathcal{E}(k) \leq D_{\max}$ . For  $n = 20$  and  $D_{\max} = 27$  there are 298 of them.
- (2) For each profile, assign the values  $\rho(1), \rho(2), \dots$  in order, choosing each displacement from the remaining multiset of the profile. At every node the term  $\sum_s h(k_s)$  is known exactly, and two further lower bounds are available: the forced inversions, meaning the inversions among assigned positions plus, for each assigned value, the number of unassigned smaller values; and the part of  $E_{\neq 1}$  coming from the assigned points alone, updated incrementally. Prune when  $\sum_s h(k_s)$  plus these two bounds exceeds the best value found so far.

The search is exhaustive for all  $\rho$  with  $D(\rho) \leq D_{\max}$ . It reproduces every known value for  $n \leq 19$ . For  $n = 20$  and  $D_{\max} = 27$  it visits  $4.7 \times 10^9$  nodes and finds minimum defect 26, attained by eight permutations. Hence  $a(20) = 190 - 26 = 164$ . One optimal permutation is the one in [Fig. 2](#):

$$\pi_{20} = (20, 18, 14, 17, 19, 13, 10, 16, 9, 15, 12, 7, 5, 11, 8, 6, 3, 4, 1, 2).$$

Its value can be checked directly: for each inverted pair  $i < j$ , normalize the triple

$$(\pi(i) - \pi(j), j - i, (\pi(i) - \pi(j))i + (j - i)\pi(i))$$

by the gcd of its entries. Distinct normalized triples correspond to distinct intersection points, and there are 164 of them. Witnesses for  $n = 17, 18, 19$  are

$$\begin{aligned} \pi_{17} &= (17, 16, 14, 15, 12, 13, 7, 10, 6, 5, 11, 9, 8, 2, 1, 4, 3), \\ \pi_{18} &= (18, 16, 12, 11, 17, 15, 14, 10, 13, 6, 9, 5, 4, 2, 8, 7, 3, 1), \\ \pi_{19} &= (19, 17, 18, 16, 12, 11, 9, 15, 7, 14, 13, 8, 10, 4, 6, 2, 5, 1, 3). \end{aligned}$$

## ACKNOWLEDGEMENTS

An earlier version of this paper (April 2026) was developed with the help of GPT-5.5 Pro. The present version corrects and extends it. The proofs have been checked against the numerical computations reported here but have not been refereed.

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